# SUMS OF TWO SQUARES WITH FEW PRIME FACTORS, II: THE MEAN SQUARE OF THE CUSPIDAL ERROR OF $\theta^2(mz)\theta^2(m'z)$, ALMOST ALL $N$, AND THE SIZE OF THE PETERSSON NORM

RUQING CHEN

ABSTRACT. Let $m, m'$ be coprime odd integers, $\Theta_{m,m'}(z) = \theta^2(mz)\theta^2(m'z)$, $G_{m,m'}$ its cuspidal part and $L = 4mm'$, so that $a_G(N) = T(m, m'; N) - M(m, m'; N)$ is the error term in the number of representations of $N$ by $m(x^2+y^2)+m'(z^2+w^2)$. Part I proved $\|G_{m,m'}\|^2 \ll_\varepsilon (mm')^\varepsilon$ for the Petersson norm and deduced the level of distribution $1/2$ for the weighted sequence $r(n)r(N-n)$ and the bounds $\Omega(n(N-n)) \le 8$, $\max(\Omega(n), \Omega(N-n)) \le 5$ for every large $N \equiv 2 \pmod 4$. Here we prove a mean-square bound $\sum_{N\le Y} |a_G(N)|^2 \le 4\pi e^2 Y(1 + 54Y/L)\|G\|^2$ for every $G \in S_2(\Gamma_0(L))$ by a covering argument; for $G_{m,m'}$ it says that the root mean square of $(T - M)/\sqrt{N}$ is $\ll (mm')^{-1/2+\varepsilon}$, and it gives the level of distribution $1 - \varepsilon$ for the weighted sequence for all but $O(X^{1-\delta})$ integers $N \le X$, and for those $N$ the bounds 5 and 4 (a result which, as an arithmetic statement about almost all $N$, is superseded by the circle method, which gives $N = p + q$ with $p \equiv q \equiv 1 \pmod 4$). We reformulate the uniform bound of Part I in terms of Poincaré series, state the individual bound $|T - M| \ll N^{1/2+\varepsilon}(mm')^{-1/2+\varepsilon}$ suggested by the data, and show that any power saving in $mm'$ for individual $N$ raises the level of distribution for every $N$ beyond $1/2$. Finally we prove the lower bound $\|G_{m,m'}\|^2 \ge \frac{1}{\pi}\big((\log m)^2 + (\log m')^2\big) - O\big(\log(2mm')(\log\log 3mm')^C\big)$ by integrating $|G|^2$ over the parts of the fundamental domain above the four Atkin–Lehner cusps at which $G_{m,m'}$ is transformed into $G_{1,mm'}$, whose first $mm'$ coefficients are $r(n) - M(1, mm'; n)$; so the norm is of the order $(\log mm')^2$ up to the factor $(mm')^\varepsilon$, and the exponent $\varepsilon$ cannot be replaced by 0. Numerically, the regions above all the cusps account for 93–98% of the norm, which is a quadratic form in $\log m$, $\log m'$ plus contributions of the other divisors of $mm'$ rather than a constant times $(\log mm')^2$; there is an exact check at the level 20, and the supremum of $|T - M|/\sqrt{N}$ over $N$ exceeds the root mean square by a bounded factor at every level computed.

## 1. RESULTS

We keep the notation of [3], referred to as Part I. Thus $r(n)$ is the number of representations of $n$ as a sum of two squares, $m, m'$ are coprime odd positive integers,

$$T(m, m'; N) = \sum_{ma+m'b=N} r(a)r(b), \qquad L = 4mm',$$

$$\Theta_{m,m'}(z) = \theta^2(mz)\theta^2(m'z) = \sum_{N\ge 0} T(m, m'; N)e(Nz) \in M_2(\Gamma_0(L)),$$

$\Theta_{m,m'} = E + G$ is the decomposition into the Eisenstein part, whose coefficients are the Siegel main terms $M(m, m'; N)$, and the cuspidal part $G = G_{m,m'} \in S_2(\Gamma_0(L))$, so that

$$a_G(N) = T(m, m'; N) - M(m, m'; N) \qquad \text{for every } N \ge 1.$$

The Petersson norm is $\|G\|^2 = \int_{\Gamma_0(L)\backslash\mathbb{H}} |G|^2 \,\mathrm{d}x\,\mathrm{d}y$. Lemma 2.1 of Part I, which is Waibel's theorem [7, Theorem 1] for the form $Q_{m,m'} = m(x^2+y^2)+m'(z^2+w^2)$, gives $\|G\| \ll (mm')^{1/4+\varepsilon}$, and the first two versions of Part I deduced from it, through a Hecke operator, the uniform bound $a_G(N) \ll d(N)N^{1/2}(mm')^{1/4+\varepsilon}$ for $(N, mm') = 1$, $v_2(N) \le 1$, the level of distribution

*Date*: October 24, 2026 (draft).

2/5 and the bounds 9 and 6, and conjectured (Conjecture 5.1 there) that the factor $(mm')^{1/4}$ can be removed. Version 3 of Part I, which this paper accompanies, proves that the norm itself is much smaller than Waibel's general bound, by computing the expansion of $\Theta_{m,m'}$ at every cusp of $\Gamma_0(L)$ with Waibel's method, and draws the consequences. We recall what we use.

**Theorem 1.1** (Part I, Theorems 1.2 and 2.3)**.** *Let $m, m'$ be coprime odd positive integers.*

(i) $\|G_{m,m'}\|^2 \ll_\varepsilon (mm')^\varepsilon$.

(ii) *For $N \geq 1$ with $(N, mm') = 1$ and $v_2(N) \leq 1$,*

$$T(m,m';N) = M(m,m';N) + O_\varepsilon\big(d(N)\, N^{1/2}(mm')^\varepsilon\big);$$

*for cube-free $m, m'$ and every $N$ with $v_2(N) \leq 1$ the error term is $O_\varepsilon(d(N)^4 N^{1/2}(mm')^\varepsilon)$ (Part I, Corollary 2.7).*

Part I deduces from (ii) that $r(n)r(N-n)$ has level of distribution 1/2 (Theorem 3.2 there) and, with sieve parameters certified in interval arithmetic, that for every sufficiently large $N \equiv 2 \pmod 4$

$$\#\big\{1 \leq n < N :\ n = x^2+y^2,\ N-n = z^2+w^2,\ \Omega(n(N-n)) \leq 8,\ \max(\Omega(n), \Omega(N-n)) \leq 5\big\}$$
$$\geq\ 0.23\, \mathfrak{S}_r(N)\, \frac{N}{(\log N)^2}, \qquad \mathfrak{S}_r(N) = \prod_{\substack{p \mid N \\ p>2}} \frac{p-1}{p-2},$$

(Theorem 1.1 there; versions 1–2 had 9 and 6). The proof of (i) shows that the expansion of $\Theta_{m,m'}$ at every cusp is again a theta series of a form $A(x_1^2+x_2^2)+B(x_3^2+x_4^2)$ with $AB \asymp mm'$, whose coefficients are $O(n^\varepsilon)$ for $n \ll mm'$ because $A\alpha + B\beta = n$ has $O(1)$ solutions, and suggests $\|G_{m,m'}\|^2 \asymp (\log mm')^2$ (Part I, Remark 2.13); numerically, $\|G_{m,m'}\|^2 \approx 0.37\,(\log mm')^2$ (Section 5). Theorem 1.5 below proves the lower bound, and Section 6 decomposes the norm over the cusps.

The first theme of this paper is the average over $N$ of the error term, which is of independent interest and gives a stronger level of distribution for almost all $N$.

**Theorem 1.2** (Mean square)**.** *Let $L \geq 1$, $G \in S_2(\Gamma_0(L))$ and $Y \geq 1$. Then*

$$\sum_{n \leq Y} |a_G(n)|^2 \ \leq\ 4\pi e^2\, Y\Big(1 + \frac{54\,Y}{L}\Big)\|G\|^2. \tag{1}$$

*In particular, for coprime odd $m, m'$, $G = G_{m,m'}$ and $Y \geq mm'$, by Theorem 1.1(i),*

$$\sum_{N \leq Y} \big|T(m,m';N) - M(m,m';N)\big|^2 \ \ll_\varepsilon\ Y^2\,(mm')^{-1+\varepsilon}, \tag{2}$$

*and hence, since $\sum_{Y/2<N\leq Y} |a_G(N)|^2/N \leq (2/Y)\sum_{N\leq Y}|a_G(N)|^2$, the root mean square of $(T-M)/\sqrt{N}$ over $Y/2 < N \leq Y$ is $\ll (mm')^{-1/2+\varepsilon}$.*

The proof of (1) is a classical covering argument and does not use the Petersson formula; the dependence on the level is exactly the factor $Y/L$, i.e. the volume of $\Gamma_0(L)\backslash\mathbb{H}$. The bound (2) is the exact order of magnitude: by the Rankin–Selberg method the mean square of $(T-M)/\sqrt{N}$ is asymptotic to $16\pi^2\|G\|^2/\operatorname{vol}(\Gamma_0(L)\backslash\mathbb{H})$, and Table 1 shows the data.

Averaging over $N$ is natural for the arithmetic application, since the sieve of Part I is applied to one $N$ at a time and an exceptional set of $N$ can be tolerated. Let, as in Part I, $N \equiv 2$

(mod 4), $\mathcal{A}=\{n\le N:\ (n,N)=1\}$, $w(n)=r(n)r(N-n)$, and for coprime square-free $d,e$ with $(de,2N)=1$ let $R(d,e)=R(d,e;N)$ be the remainder in

$$\sum_{\substack{n\in\mathcal{A}\\ d\mid n,\ e\mid N-n}} w(n)=X_N\,h(d)h(e)+R(d,e;N)$$

(this is $S(d,e)=M(d,e)+R(d,e)$ of Part I, Section 3; we write $X_N$ for the main term $X$ there, to avoid a clash with the length $X$ of the range of $N$ below).

**Theorem 1.3** (Level of distribution $1-\varepsilon$ for almost all $N$)**.** *Let $0<\varepsilon<1/20$, $X\ge X_0(\varepsilon)$ and $D=X^{1-3\varepsilon}$. Then*

$$\sum_{\substack{X/2<N\le X\\ N\equiv 2\ (4)}}\ \sum_{\substack{de\le D,\ (d,e)=1\\ (de,2N)=1}} \mu^2(de)\,d_3(de)\,|R(d,e;N)| \ \ll_\varepsilon\ X^{2-\varepsilon/2}.$$

*Consequently, for all but $O_\varepsilon(X^{1-\varepsilon/4})$ integers $N\equiv 2\ (\mathrm{mod}\ 4)$ in $(X/2,X]$,*

$$\sum_{\substack{de\le N^{1-3\varepsilon}\\ (d,e)=1,\ (de,2N)=1}} \mu^2(de)\,d_3(de)\,|R(d,e;N)| \ \le\ N^{1-\varepsilon/5}. \tag{3}$$

The sieve of Part I, Section 4, uses the level of distribution only through (3); running it at the level $\theta=19/20$ gives the following.

**Corollary 1.4.** *For all but $O(X^{1-\delta})$ integers $N\equiv 2\ (\mathrm{mod}\ 4)$ in $(X/2,X]$, with an absolute $\delta>0$,*

$$\#\big\{1\le n<N:\ n=x^2+y^2,\ N-n=z^2+w^2,\ \Omega(n(N-n))\le 5,\ \max(\Omega(n),\Omega(N-n))\le 4\big\} \ge\ 0.34\,\mathfrak{S}_r(N)\,\frac{N}{(\log N)^2},$$

*the integers counted being coprime to $N$ and products of distinct primes $\equiv 1\ (\mathrm{mod}\ 4)$ exceeding $N^{2/25}$.*

The certified sieve parameters are $\alpha=2/25$, $\lambda=1/3$, $\beta_1=69/100$ (Section 3). The pairs $(5,4)$ for almost all $N$ and $(8,5)$ for every $N$ should be compared with $(9,6)$ for every $N$, which is what Waibel's bound alone gives (Part I, Remark 4.5). A floating-point scan (Figure 2) shows that the best attainable value of $1/\lambda+1/\beta_1$ drops below 4 only for $\theta>0.88$; at $\theta=19/20$ the scan gives 3.83 (at $\alpha=0.10$, $\beta_1=0.47$, $\lambda=0.59$, with $1/\lambda+2/\beta_1=5.96$), i.e. the pair $(5,3)$, which we have not certified, for the reason explained next.

Corollary 1.4 must also be compared with what the circle method gives for almost all $N$. By the method of Montgomery and Vaughan [5], carried out with the characters modulo 4 (see [2] for Goldbach's problem with primes in prescribed residue classes), all but $O(X^{1-\delta})$ integers $N\equiv 2$ (mod 4) up to $X$ are sums $p+q$ of two primes $p\equiv q\equiv 1\ (\mathrm{mod}\ 4)$, with $\gg \mathfrak{S}_r(N)N/(\log N)^2$ representations; such primes are sums of two squares, so for almost all $N$ the pair $(1,1)$ holds with the count of Corollary 1.4. As an arithmetic statement about almost all $N$, Corollary 1.4 is therefore far from new. What is new is Theorem 1.3, a level of distribution $1-\varepsilon$ for the weighted sequence $r(n)r(N-n)$, together with the observation that the sieve of Part I extracts $(5,4)$ from it with no further input; the sieve route adds something only for *every* $N$, where the circle method gives nothing (Goldbach's problem) and where Part I gives $(8,5)$. Section 4 formulates the individual bound that would raise the level to $1-\varepsilon$ for every $N$ and with it give $(5,4)$ for every $N$ (Proposition 4.2).

The third theme is the size of the norm itself. Theorem 1.1(i) is an upper bound. The proof in Part I shows that the expansion of $\Theta_{m,m'}$ at every cusp of $\Gamma_0(L)$ is a theta series of a form $A(x_1^2+x_2^2)+B(x_3^2+x_4^2)$ with $AB \asymp mm'$; at the four cusps which are the images of $\infty$ under the Atkin–Lehner involutions $W_m$, $W_{4m}$, $W_{m'}$, $W_{4m'}$ this form is $x_1^2+x_2^2+mm'(x_3^2+x_4^2)$, whose first $mm'$ representation numbers are the $r(n)$ themselves, and there $G_{m,m'}$ is transformed, exactly, into $G_{1,mm'}$. Integrating $|G|^2$ over the parts of the fundamental domain above these four cusps gives a lower bound.

**Theorem 1.5** (Lower bound for the norm)**.** *Let $m, m'$ be coprime odd positive integers with $mm' > 1$, and let $a(n) = a_{G_{1,mm'}}(n) = T(1,mm';n) - M(1,mm';n)$, so that $a(n) = r(n) - M(1,mm';n)$ for $1 \le n < mm'$.*

(i) $$\|G_{m,m'}\|^2 \;\ge\; \frac{1}{4\pi} \sum_{Q\in\{m,\,4m,\,m',\,4m'\}} \sum_{n\ge1} \frac{|a(n)|^2}{n}\, e^{-4\pi n/Q}.$$

(ii) *There is an absolute constant $C$ such that*

$$\begin{aligned}\|G_{m,m'}\|^2 &\ge \frac{1}{\pi}\big((\log m)^2+(\log m')^2\big) - C\log(2mm')\,(\log\log 3mm')^C\\ &\ge \frac{1}{2\pi}(\log mm')^2 - C\log(2mm')\,(\log\log 3mm')^C.\end{aligned}$$

Together with Theorem 1.1(i), $(\log mm')^2 \ll \|G_{m,m'}\|^2 \ll_\varepsilon (mm')^\varepsilon$ for $mm' \ge C'$, and the exponent $\varepsilon$ cannot be replaced by 0. The inequality (i) is nearly an equality when $m = 1$: the four regions then cover the fraction $5/(2\pi) = 0.80$ of the area of $\Gamma_0(L)\backslash\mathbb{H}$, and numerically they carry 98% of the norm (Table 3). The proof is in Section 6, which also computes, for square-free $m, m'$, the contribution of every cusp and explains the empirical law of Section 5: the norm is not $c(\log mm')^2$ but $\frac{1}{\pi}((\log m)^2+(\log m')^2) + \frac{5}{\pi^2}\log m\log m' + \cdots$, a sum over the divisors of $mm'$.

Section 4 restates the uniform bound of Theorem 1.1(ii) in terms of Poincaré series and formulates the individual bound, Section 5 contains the numerical evidence: the data of Part I, an exact check of the whole numerical chain at the level 20, the empirical law $\|G_{m,m'}\|^2 \approx 0.37(\log mm')^2$, which corrects a remark in the first two versions of Part I, and the supremum of $|T-M|/\sqrt{N}$ over $N$; and Section 6 proves Theorem 1.5 and decomposes the norm over the cusps.

## 2. Proof of Theorem 1.2

Let $G(z) = \sum_{n\ge1} a(n)e(nz) \in S_2(\Gamma_0(L))$ and put $F(z) = y^2|G(z)|^2$, $z = x+iy$. Since $G(\gamma z) = (cz+d)^2 G(z)$ and $\operatorname{Im}(\gamma z) = y/|cz+d|^2$ for $\gamma = \left(\begin{smallmatrix} a & b \\ c & d\end{smallmatrix}\right) \in \Gamma_0(L)$, the function $F$ is $\Gamma_0(L)$-invariant, and with the invariant measure $\mathrm{d}\mu = y^{-2}\mathrm{d}x\,\mathrm{d}y$ we have $\|G\|^2 = \int_{\mathcal F} F\,\mathrm{d}\mu$ for any fundamental domain $\mathcal F$ of $\Gamma_0(L)$.

Let $y_0 = 1/(4\pi Y)$ and $R = \{x+iy:\ 0\le x<1,\ y_0 \le y \le 2y_0\}$ (half-open in $x$, so that each orbit of the translations $z\mapsto z+b$ meets $R$ at most once). By Parseval, $\int_0^1 |G(x+iy)|^2\,\mathrm{d}x = \sum_n |a(n)|^2 e^{-4\pi n y}$, hence

$$\int_R |G|^2\,\mathrm{d}x\,\mathrm{d}y = \sum_n |a(n)|^2 \int_{y_0}^{2y_0} e^{-4\pi n y}\,\mathrm{d}y \;\ge\; \frac{e^{-2}}{4\pi Y}\sum_{n\le Y}|a(n)|^2,$$

because $4\pi n y \le 8\pi Y y_0 = 2$ for $n \le Y$, $y \le 2y_0$. On the other hand $\int_R |G|^2\,\mathrm{d}x\,\mathrm{d}y = \int_R F\,\mathrm{d}\mu$, and

$$\int_R F\,\mathrm{d}\mu = \sum_{\gamma\in\Gamma_0(L)/\{\pm1\}} \int_{R\cap\gamma\mathcal F} F\,\mathrm{d}\mu = \int_{\mathcal F} F(w)\,\#\{\gamma:\ \gamma w\in R\}\,\mathrm{d}\mu(w) \;\le\; \mu_R\,\|G\|^2,$$

where $\mu_R = \sup_{z\in R} \#\{\gamma \in \Gamma_0(L)/\{\pm1\} :\ \gamma z \in R\}$ (if $\#\{\gamma : \gamma w \in R\} \neq 0$ for some $w \in \mathcal{F}$, pick $\gamma_0$ with $z = \gamma_0 w \in R$; then $\gamma \mapsto \gamma\gamma_0^{-1}$ is a bijection from $\{\gamma : \gamma w \in R\}$ onto $\{\gamma : \gamma z \in R\}$, so the count at $w$ is at most $\mu_R$). It remains to bound $\mu_R$. Fix $z = x+iy \in R$ and let $\gamma = \left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right) \in \Gamma_0(L)$ with $\gamma z \in R$.

- If $c = 0$, then $\gamma = \pm \left(\begin{smallmatrix} 1 & b \\ 0 & 1 \end{smallmatrix}\right)$ and $\gamma z = z + b$, so $x + b \in [0, 1)$ forces $b = 0$: one element.
- If $c \neq 0$, we may take $c > 0$, since $\gamma$ and $-\gamma$ are the same element of $\Gamma_0(L)/\{\pm1\}$. Then $L \mid c$, and $\mathrm{Im}(\gamma z) = y/|cz+d|^2 \in [y_0, 2y_0]$ together with $y \in [y_0, 2y_0]$ gives $|cz+d|^2 \in [1/2, 2]$. Since $|cz+d|^2 = (cx+d)^2+c^2y^2$, this forces $c^2y^2 \le 2$, hence $c \le \sqrt{2}/y_0 = 4\sqrt{2}\,\pi Y$, and $|cx+d| \le \sqrt{2}$. The number of positive multiples $c$ of $L$ with $c \le 4\sqrt{2}\pi Y$ is $\lfloor 4\sqrt{2}\pi Y/L \rfloor \le 4\sqrt{2}\pi\, Y/L$; for each $c$ there are at most 3 integers $d$ with $|cx + d| \le \sqrt{2}$; and for each pair $(c, d)$ the elements of $\Gamma_0(L)$ with bottom row $(c, d)$ are $\left(\begin{smallmatrix} 1 & k \\ 0 & 1 \end{smallmatrix}\right)\gamma_0$, $k \in \mathbb{Z}$, whose images $\gamma_0 z + k$ have real part in $[0, 1)$ for exactly one $k$. This gives at most $4\sqrt{2}\pi \cdot 3\, Y/L = 12\sqrt{2}\pi\, Y/L < 54\, Y/L$ elements.

Hence $\mu_R \le 1 + 54Y/L$, and

$$\sum_{n\le Y} |a(n)|^2 \le 4\pi e^2 Y \int_R F \,\mathrm{d}\mu \le 4\pi e^2\, Y\Big(1 + \frac{54Y}{L}\Big)\|G\|^2,$$

which is (1). For $G = G_{m,m'}$, $L = 4mm'$ and $Y \ge mm'$ the bracket is $\le 1 + 54Y/(4mm') \le 15Y/(mm')$, and Theorem 1.1(i) gives (2). □

*Remark* 2.1. The argument is the classical one for bounding $\sum_{n\le Y} |a(n)|^2$ by the norm (compare [6, Chapter 5]); its only input is the invariance of $y^2|G|^2$. The strip $y_0 \le y \le 2y_0$ with $y_0 = 1/(4\pi Y)$ is chosen so that the exponential factor in Parseval's identity stays at $e^{-2}$ for all $n \le Y$. For $Y \ge L$ the factor $Y^2/L$ in (1) is of the correct order of magnitude: by the Rankin–Selberg method, $\sum_{n\le Y} |a(n)|^2/n \sim 16\pi^2\|G\|^2 Y/\operatorname{vol}(\Gamma_0(L)\backslash\mathbb{H})$ with $\operatorname{vol} = \frac{\pi}{3} L \prod_{p\mid L}(1 + 1/p)$ (measure $\mathrm{d}\mu$), so that $\sum_{n\le Y} |a(n)|^2 \sim 8\pi^2\|G\|^2 Y^2/\operatorname{vol}$ for fixed $G$ as $Y \to \infty$. Theorem 1.2 makes this uniform in the level, at the cost of the constant.

## 3. Proof of Theorem 1.3 and Corollary 1.4

3.1. **The remainder on average.** We follow the proof of Theorem 3.2 of Part I, replacing the uniform bound for $|T - M|$ by Theorem 1.2 after an application of Cauchy–Schwarz over $N$. Recall from Part I, (16), that for $N \equiv 2 \pmod 4$ and coprime square-free $d, e$ with $(de, 2N) = 1$,

$$(4)\quad R(d, e; N) = \sum_{t\mid N^\flat} \mu(t) \sum_{m\mid d^2} \sum_{m'\mid e^2} c_d(m) c_e(m') \sum_{\substack{t_1\mid t^2 \\ t\mid t_1}} c_t(t_1)\,\big(a_G(N) - a_G(N/2)\big), \qquad G = G_{t_1 m, m'},$$

where $N^\flat$ is the product of the odd primes dividing $N$, $m = ds$, $m' = es'$ with $s \mid d$, $s' \mid e$, and likewise $t_1 = tu$ with $u \mid t$ (the coefficients $c_d(m)$, $c_t(t_1)$ of Part I, (15), vanish unless $m = ds$ with $s \mid d$, resp. $t_1 = tu$ with $u \mid t$; in particular $t \mid t_1$); the parameters $t_1 m, m'$ of the form $Q_{t_1 m, m'}$ are coprime, odd and cube-free, $\tilde{m}\tilde{m}' := t_1 mm' = t_1 ss' \cdot de$, and the coefficients $c$ satisfy $|c_d(m)c_e(m')c_t(t_1)| \le 3^{\omega(d)+\omega(e)+\omega(t)}$. (For the reader's convenience: $c_d$ is the multiplicative coefficient of Part I, (15), with $c_p(p) = 2$, $c_p(p^2) = -1$ for a split prime $p$ and $c_q(q) = 0$, $c_q(q^2) = 1$ for an inert prime $q$; it encodes the identities $r(n) = 2r(n/p) - r(n/p^2)$ and $r(n) = r(n/q^2)$, which turn the condition $d \mid n$ into a combination of theta series $\theta^2(mz)$. The sum over $t$ is the Möbius inversion of the condition $(n, N) = 1$, and $a_G(N) - a_G(N/2)$ selects the odd $n$.) Let $\eta = \varepsilon/10$. For $N \le X$ with $X \ge X_0(\varepsilon)$ we have, as in (F1)–(F3) of Part I: $d_3(de)\, 3^{\omega(d)+\omega(e)+\omega(t)}\, 4^{\omega(N)} d(N)^2 \le X^\eta$; for fixed $(d, e, N)$ the number of tuples $(s, s', t, t_1)$ is at

most $X^\eta$; and $\sum_{k\le D} d(k)/k \le X^\eta$, $\sum_{k\le D} d(k)k^{-1/2} \le D^{1/2}X^\eta$. We also use the two bounds for a single term,

$$|a_G(N')| \le T(\tilde m,\tilde m';N') + M(\tilde m,\tilde m';N') \ll X^{2\eta}\Big(\frac{X}{\tilde m\tilde m'}+1\Big) \qquad (N'\le X), \tag{5}$$

$$\sum_{n\le X}|a_G(n)| \le X^{1/2}\Big(\sum_{n\le X}|a_G(n)|^2\Big)^{1/2} \ll X\Big(1+\frac{X}{L}\Big)^{1/2}L^\eta, \tag{6}$$

where $G = G_{\tilde m,\tilde m'}$ and $L = 4\tilde m\tilde m'$, the first being (21) of Part I, the second Cauchy–Schwarz, (1) and Theorem 1.1(i); for $L \le X$ the right-hand side of (6) is $\ll X^{3/2}(\tilde m\tilde m')^{-1/2+\eta}$. Let $\Sigma$ denote the double sum in Theorem 1.3. Inserting (4) and splitting the tuples according to the size of $t_1ss'$:

*Tuples with $t_1ss' > X^\varepsilon$.* For each $N$ and each tuple, (5) with $\tilde m\tilde m' > X^\varepsilon de$ gives $|a_G(N')| \ll X^{2\eta}(X^{1-\varepsilon}/(de)+1)$. Summing over the $\le X^\eta$ tuples, over $N\le X$ and over $de\le D$ with the weights $d_3(de)\cdot 3^\omega \le X^\eta$, these tuples contribute

$$\ll X^{1+4\eta}\sum_{k\le D} d(k)\Big(\frac{X^{1-\varepsilon}}{k}+1\Big) \ll X^{2-\varepsilon+5\eta} + X^{1+5\eta}D \ll X^{2-\varepsilon/2}.$$

*Tuples with $t_1ss' \le X^\varepsilon$.* Now $t \le t_1 \le X^\varepsilon$ (as $t \mid t_1$), so $t$ runs over at most $X^\varepsilon$ values, $t_1$ over at most $d(t^2) \le X^\eta$ values, and $(s,s')$ over at most $d(d)d(e) \le X^\eta$ values; the forms $G = G_{t_1m,m'}$ have level $L = 4t_1ss'de \le 4X^\varepsilon D \le X$, so that the right-hand side of (6) is $\ll X^{3/2}(\tilde m\tilde m')^{-1/2+\eta}$. For a fixed tuple and fixed $(d,e)$, we drop the condition $t \mid N$ except for the count: the number of $N\le X$ with $t\mid N$ is $\le X/t$, and Cauchy–Schwarz gives

$$\sum_{\substack{N\le X\\ t\mid N}}\big(|a_G(N)|+|a_G(N/2)|\big) \le 2\Big(\frac{X}{t}\Big)^{1/2}\Big(\sum_{n\le X}|a_G(n)|^2\Big)^{1/2}$$
$$\ll \Big(\frac{X}{t}\Big)^{1/2} X\,(\tilde m\tilde m')^{-1/2+\eta} \le X^{3/2+\eta}\,t^{-1/2}(de)^{-1/2},$$

using (1) in the form $\sum_{n\le X}|a_G(n)|^2 \ll X^2\|G\|^2/L$ and $(\tilde m\tilde m')^{-1/2+\eta} \le X^\eta(de)^{-1/2}$. Summing over $t\le X^\varepsilon$ ($\sum_t t^{-1/2} \le 2X^{\varepsilon/2}$), over the other tuple variables ($\le X^{2\eta}$), and over $de\le D$ with the weights $\le X^\eta$,

$$\ll X^{3/2+4\eta+\varepsilon/2}\sum_{k\le D} d(k)k^{-1/2} \ll X^{3/2+5\eta+\varepsilon/2}D^{1/2} = X^{2-\varepsilon+5\eta} = X^{2-\varepsilon/2},$$

since $D^{1/2} = X^{1/2-3\varepsilon/2}$. Both contributions are $\ll X^{2-\varepsilon/2}$, which proves the first assertion. (The margin is small but positive: the choice $D = X^{1-3\varepsilon}$ saves $X^{-3\varepsilon/2}$ in $D^{1/2}$, of which the sum over $t$ costs $X^{\varepsilon/2}$ and the divisor-type factors cost $X^{5\eta} = X^{\varepsilon/2}$; this is why $\eta = \varepsilon/10$.)

For the second, let $S(N)$ be the inner sum with $D = X^{1-3\varepsilon}$. The number of $N\in(X/2,X]$, $N\equiv 2 \pmod 4$, with $S(N) > X^{1-\varepsilon/4}$ is at most $\Sigma/X^{1-\varepsilon/4} \ll X^{1-\varepsilon/4}$. For the remaining $N$, since $N^{1-3\varepsilon}\le D$ and $X^{1-\varepsilon/4} \le (2N)^{1-\varepsilon/4} \le N^{1-\varepsilon/5}$ for large $X$, (3) holds. □

*Remark* 3.1. The exponent 1 is the balance $X^{3/2}D^{1/2} = X^2$ between the mean-square bound $X^{3/2}(de)^{-1/2}$ for the sum over $N$ of one remainder and the number $D$ of moduli; with a norm bound $\|G_{m,m'}\| \ll (mm')^{\beta+\varepsilon}$ in place of Theorem 1.1(i) one gets the level $1/(2(1+\beta))$ for every $N$ (Part I, Remark 3.3) and $1/(1+2\beta)$ for almost all $N$; Waibel's $\beta = 1/4$ gives $2/5$ and $2/3$, and $\beta = 0$ gives $1/2$ and 1. The exceptional set $X^{1-\varepsilon/4}$ comes from Markov's inequality applied to a first moment of $S(N)$, which in turn rests on the second moment of $a_G(N)$; a bound for a higher moment would shrink it, and we do not pursue this.

3.2. **The sieve at the level** 19/20**.** Proposition 4.3 of Part I and its proof use the level of distribution only through the bound (3) (there with 2/5); they hold verbatim, for a given $N$, with $\theta = 2/5$ replaced by any $\theta \in (0,1)$ for which (3) holds with $N^{\theta-\varepsilon}$ in place of $N^{1-3\varepsilon}$, provided $\alpha < \beta_1 < \theta$. Lemmas 4.1 and 4.2 of Part I do not involve the level at all. Hence, for every $N$ satisfying (3) with $3\varepsilon < 1/20$, the conclusion of Proposition 4.3 holds with $\theta = 19/20$, and the proof of Theorem 1.1 of Part I goes through with the parameters certified for $\theta = 19/20$. The script `certify_modular.py` of Part I, extended by the argument `19/20` in the repository of this paper, certifies in interval arithmetic

$$\theta = \tfrac{19}{20}, \qquad \alpha = \tfrac{2}{25}, \qquad \lambda = \tfrac{1}{3}, \qquad \beta_1 = \tfrac{69}{100},$$
$$\tfrac{1}{\lambda} + \tfrac{2}{\beta_1} = \tfrac{407}{69} < 6, \qquad \tfrac{1}{\lambda} + \tfrac{1}{\beta_1} = \tfrac{307}{69} < 5, \qquad L - \lambda R \geq 0.085058$$

(with $L \geq 0.999484$, $R \leq 2.743278$, $\theta_A = 0.4740$; the output files are listed at the end of the paper). By Lemma 4.2 of Part I this gives $\Omega(n(N-n)) \leq 5$ and $\max(\Omega(n), \Omega(N-n)) \leq 4$, and the weight of each $n$ counted is $w(n) = 16 \cdot 2^{\Omega(n(N-n))} \leq 2^9$. As in the proof of Theorem 1.1 of Part I, the number of integers counted is at least

$$\big(0.085058 + o(1)\big)\, 2^{-9} \cdot 8\, \mathfrak{S}_r(N)\, c_{2h}\, \frac{N}{\alpha^2 (\log N)^2} \;\geq\; 0.34\, \mathfrak{S}_r(N) \frac{N}{(\log N)^2},$$

since $0.085058 \cdot 2^{-9} \cdot 8 \cdot 1.66485 \cdot (25/2)^2 = 0.3457\ldots$. The exceptional set is that of Theorem 1.3 with, say, $\varepsilon = 1/80$, so $\delta = 1/320$ is admissible. □

*Remark* 3.2. The same sieve gives $(8,5)$ at the level $1/2$ (Part I), $(6,4)$ at $2/3$ (certified: $\alpha = 3/50$, $\lambda = 1/3$, $\beta_1 = 51/100$, $L - \lambda R \geq 0.036741$) and $(5,4)$ at $19/20$. The floating-point scan of Figure 2 shows that $1/\lambda + 2/\beta_1$ drops below 6 only for $\theta > 0.83$, and that the best attainable value of $1/\lambda + 1/\beta_1$ drops below 4 for $\theta > 0.88$: at $\theta = 19/20$ the minimum is 3.83, attained at $\alpha = 0.10$, $\beta_1 = 0.47$, $\lambda = 0.59$, where $1/\lambda + 2/\beta_1 = 5.96$, so that the Richert weights give $(5,3)$ for almost all $N$ in the floating-point model, with the sifting parameter $\alpha$ at the edge of the scanned range. We have not certified this set, because Corollary 1.4 is in any case superseded for almost all $N$ by the circle method (Section 1); it is an "almost all $N$" statement with a power-saving exceptional set and says nothing about any individual $N$, which is where the sieve route matters (Section 4.1).

*Remark* 3.3 (The vector sieve is not the natural sieve here). The sieve of Part I treats the two sides $n$ and $N-n$ by the vector sieve of Brüdern and Fouvry, with two linear components of levels $N^{\theta_A}$ and $N^{\theta-\theta_A}$ and the lower bound $L = f_1(s_A)F_1(s_B) + F_1(s_A)f_1(s_B) - F_1(s_A)F_1(s_B)$. For $s_A, s_B \leq 3$ one has $f_1/F_1(s) = \log(s-1)$, so $L > 0$ if and only if $(s_A - 1)(s_B - 1) > e$, and with $s = s_A + s_B$ fixed this requires $s > 2 + 2\sqrt{e} = 5.297\ldots$, attained at the symmetric split. The product sequence $n(N-n)$, however, is a sieve problem of dimension 2 in the classical sense: under the weight $r(n)r(N-n)$ the density of $p \mid n(N-n)$ is $2h(p)$, $\sum_{p<z} 2h(p) = 2\log\log z + O(1)$, the condition $d \mid n(N-n)$ with $(n,N) = 1$ splits uniquely as $d = d_1 d_2$, $d_1 \mid n$, $d_2 \mid N-n$, and the remainder for the modulus $d$ is $\sum_{d_1 d_2 = d} R(d_1, d_2)$, which is exactly what Theorem 3.2 of Part I and Theorem 1.3 bound (this is the reason for the weight $d_3(de)$ there). The two-dimensional sieve of Diamond, Halberstam and Richert [4] applies to it directly, with the joint level $d \leq N^{\theta}$ and the sieving limit $\beta_2 = 4.2665\ldots$ in place of 5.297; its upper and lower functions $F_2, f_2$ should be used in place of the products of $F_1, f_1$ both in $L$ and in the weight term $R$. We expect this to improve the pairs $(k, k')$ at every level without any new analytic input, and we have not carried it out; the bound for $\max(\Omega(n), \Omega(N-n))$ then needs a separate argument, since the two-dimensional sieve controls only $\Omega(n(N-n))$. For comparison, for the unweighted sequence $n(N-n)$, which has level of distribution 1 trivially, the two-dimensional weighted sieve gives $N = P_2 + P_3$ for every large even $N$ (Wang Yuan, 1957).

## 4. The uniform bound and the Poincaré series

Let $L = 4mm'$ and let $P_N \in S_2(\Gamma_0(L))$, $N \geq 1$, be the Riesz representative of the $N$-th coefficient: the unique cusp form with

$$\langle f, P_N \rangle = \frac{a_f(N)}{4\pi N} \qquad \text{for all } f \in S_2(\Gamma_0(L)), \tag{7}$$

i.e. $P_N = (4\pi N)^{-1} \sum_{f \in \mathcal{B}} \overline{a_f(N)}\, f$ for an orthonormal basis $\mathcal{B}$. (This is the $N$-th Poincaré series; for weight 2 the series itself converges only conditionally, which is why we define $P_N$ by (7).) By (11) of Part I,

$$\|P_N\|^2 = \frac{a_{P_N}(N)}{4\pi N} = \frac{1}{(4\pi N)^2} \sum_{f \in \mathcal{B}} |a_f(N)|^2 = \frac{\Delta_L(N,N)}{4\pi N},$$

where $\Delta_L(N,N)$ is the Petersson density. Hence, for $G = G_{m,m'}$,

$$|a_G(N)|^2 = (4\pi N)^2 |\langle G, P_N \rangle|^2 = 4\pi N\, \Delta_L(N,N)\, \|G\|^2 \cos^2 \angle(G, P_N), \tag{8}$$

where $\cos \angle(G, P_N) = |\langle G, P_N \rangle| / (\|G\|\, \|P_N\|)$. The identity separates the three quantities that enter any bound for $a_G(N)$: the norm $\|G\|^2$, which by Theorem 1.1(i) is $\ll L^\varepsilon$; the Petersson density, which is $\leq C'$ for $N \leq L^{1/2}$ (Part I, Lemma 2.2, Step 4) and $1 + O(N^{1/2+\varepsilon}/L)$ in general; and the alignment $\cos^2 \angle(G, P_N) \leq 1$. Theorem 1.1(ii) is obtained from (8) by bounding the alignment trivially: after the coprime part of $N$ is moved into a Hecke operator (Part I, Lemma 2.2), only $P_{n_2}$ with $n_2 \in \{1, 2\}$ occurs, where the density is bounded, and the norm does the rest. Theorem 1.2 says what the alignment is on average: inserting (8) into (1) and restricting to $N \in (Y/2, Y]$,

$$\frac{1}{Y/2} \sum_{Y/2 < N \leq Y} \Delta_L(N,N) \cos^2 \angle(G, P_N) \;\leq\; 4e^2 \Big( \frac{1}{Y} + \frac{54}{L} \Big) \;\ll\; \frac{1}{L} \qquad (Y \geq L), \tag{9}$$

the value of $\cos^2$ for a unit vector in general position in the $\asymp L$-dimensional space $S_2(\Gamma_0(L))$: on average over $N$, the theta series is no more aligned with the Poincaré series than a random cusp form of level $L$ would be, and Theorem 1.1(ii) says that no single $N$ is aligned by more than $N^\varepsilon L / \|G\|^2 \gg L^{1-\varepsilon}$ times the average. Expanding in an orthogonal basis $\mathcal{B}^*$ of Hecke eigenforms as in Part I, Section 5.5,

$$\langle G, P_N \rangle = \sum_{\varphi \in \mathcal{B}^*} \frac{\langle G, \varphi \rangle \langle \varphi, P_N \rangle}{\|\varphi\|^2} = \frac{1}{4\pi N} \sum_{\varphi \in \mathcal{B}^*} \frac{\langle \Theta, \varphi \rangle\, a_\varphi(N)}{\|\varphi\|^2},$$

so that Theorem 1.1(ii) is the bound $\ll N^{1/2+\varepsilon}$ for this spectral sum, uniformly in $m, m'$; the first two versions of Part I asked for a proof through the cancellation between the coefficients $\langle \Theta, \varphi \rangle$ and $a_\varphi(N)$. The proof in Part I does not see this cancellation: it shows instead that $\sum_\varphi |\langle \Theta, \varphi \rangle|^2 / \|\varphi\|^2 = \|G\|^2$ is itself small, i.e. that the theta series has a small cuspidal projection to begin with, and the spectral sum is then bounded by Cauchy–Schwarz.

### 4.1. The individual bound.

Theorem 1.2 and the data (Section 5.4) say that the cancellation which the proof does not see is there for most $N$, and the sieve for every $N$ would profit from it for each $N$. We state the strongest form suggested by the data.

*Conjecture* 4.1. For coprime odd $m, m'$ and $N \geq 1$ with $(N, mm') = 1$ and $v_2(N) \leq 1$,

$$|T(m,m';N) - M(m,m';N)| = |a_{G_{m,m'}}(N)| \;\ll_\varepsilon\; N^{1/2+\varepsilon} (mm')^{-1/2+\varepsilon}.$$

In the terms of (8): $\Delta_L(N,N) \cos^2 \angle(G, P_N) \ll N^\varepsilon L^{-1+\varepsilon} / \|G\|^2$, i.e. no Poincaré series $P_N$ is aligned with $G$ by more than $N^\varepsilon L^\varepsilon$ times the average (9).

Conjecture 4.1 is the mean-square bound (2) applied to each $N$ separately; it is consistent with the trivial lower bound $|a_G(N)| \geq \operatorname{dist}(M, 8\mathbb{Z})$ for $N \ll L$, where $M \asymp N/(mm') \ll 1$, and with the data of Section 5.4, where the supremum over $10^6$ to $4 \cdot 10^6$ values of $N$ exceeds the root mean square by a factor between 5 and 10 at every level computed. We do not know how to prove any power saving in $mm'$ for an individual $N$. In the spectral form the question is a twisted first moment $\sum_\varphi \langle \Theta, \varphi \rangle a_\varphi(N)/\|\varphi\|^2$ over the forms of level $L$ with the "long" twist $N \geq L$ (the sieve at the level $\theta$ uses $N \approx L^{1/\theta}$), outside the range $N \leq L^{1-\delta}$ in which the Petersson formula evaluates such moments. Any saving would raise the level for every $N$:

**Proposition 4.2.** *Let $0 \leq \eta \leq \frac{1}{2}$ and suppose that $|a_{G_{m,m'}}(N)| \ll_\varepsilon N^{1/2+\varepsilon}(mm')^{-\eta+\varepsilon}$ for all coprime odd $m, m'$ and all $N$ with $(N, mm') = 1$, $v_2(N) \leq 1$. Then Theorem 3.2 of Part I holds for every $N \equiv 2 \pmod 4$ with $D = N^{\theta-\varepsilon}$, $\theta = 1/(2(1-\eta))$, i.e. $r(n)r(N-n)$ has level of distribution $1/(2(1-\eta))$. In particular, Conjecture 4.1 gives the level $1-\varepsilon$ for every $N$, and then the sieve of Section 3 gives, for every sufficiently large $N \equiv 2 \pmod 4$, $\Omega(n(N-n)) \leq 5$ and $\max(\Omega(n), \Omega(N-n)) \leq 4$ for at least $0.34\,\mathfrak{S}_r(N)N/(\log N)^2$ integers $n$.*

*Proof.* We follow the proof of Theorem 3.2 of Part I and only change the bound for one remainder. Each term of $R(d,e)$ is $\pm c\, a_G(N')$ for a form $G = G_{\tilde m, \tilde m'}$ with $\tilde m \tilde m' = t_1 s s' \cdot de$, where $t \mid N^\flat$, $t_1 \mid t^2$, and $N' \in \{N, N/2\}$ may have common factors with $\tilde m \tilde m'$ at the primes of $t$ only. Lemma 2.5 of Part I reduces $a_G(N')$ to $\leq 4^{\omega(N)} d(N)$ terms $a_{G_{m_j, m_j'}}(N_j)$ with $(N_j, m_j m_j') = 1$, $N_j \mid N'$ and $m_j m_j' \geq \tilde m \tilde m'/t_1^2 \geq de/t_1^2$ (each reduction step at a prime $p$ divides the product of the coefficients by at most $p^2$, and only the primes of $t_1$ are involved). Hence, by the hypothesis with $\varepsilon/20$ in place of $\varepsilon$,

$$|a_G(N')| \ll 4^{\omega(N)} d(N)\, N^{1/2+\varepsilon/20}\, (de)^{-\eta}\, t_1^{2\eta}\, (mm')^{\varepsilon/20}.$$

Put $\eta' = \varepsilon/20$. For the terms with $t_1 s s' \leq N^\varepsilon$ this is $\ll N^{1/2+3\eta'+2\varepsilon\eta}(de)^{-\eta}$ by (F1) of Part I, and their total is

$$\ll N^{1/2+5\eta'+\varepsilon} \sum_{k \leq D} d(k) k^{-\eta} \ll N^{1/2+6\eta'+\varepsilon} D^{1-\eta} = N^{1/2+6\eta'+\varepsilon} N^{1/2-\varepsilon(1-\eta)} \leq N^{1-\varepsilon/2+6\eta'},$$

since $D^{1-\eta} = N^{(\theta-\varepsilon)(1-\eta)} = N^{1/2-\varepsilon(1-\eta)}$ and $1-\eta \geq \frac{1}{2}$. The terms with $t_1 s s' > N^\varepsilon$ are bounded by the trivial bound (5) exactly as in Part I, by $N^{1-\varepsilon+5\eta'} + N^{\theta+5\eta'}$. With $\theta \leq 1-\varepsilon$ (replace $\varepsilon$ by $\min(\varepsilon, 1-\theta)$ if necessary) the total is $\ll N^{1-\varepsilon/2+6\eta'} \leq N^{1-\varepsilon/5}$, which is the conclusion of Theorem 3.2 of Part I with $N^{1-\varepsilon/5}$ in place of $N^{1-\varepsilon/3}$; Proposition 4.3 of Part I needs only a power saving. The last assertion is Section 3, where the parameters at the level 19/20 were certified for any $N$ satisfying (3). □

The exponent $\theta = 1/(2(1-\eta))$ interpolates between the level 1/2 of Part I ($\eta = 0$) and the level 1 ($\eta = \frac{1}{2}$); the level 2/3, which gives (6, 4) for every $N$ with the certified parameters of Section 3, needs only $\eta = \frac{1}{4}$, i.e. a saving of $(mm')^{-1/4}$ over Theorem 1.1(ii), half of what the mean square (2) gives on average.

## 5. Numerical evidence

5.1. **The mean square and the norm.** The script `mean_square_check.py` computes, with the tables and the FFT convolution of `sup_error_scan.py` of Part I, the sums $S_2(Y) = \sum_{N \in \mathcal{N}_Y} |a_G(N)|^2$ over the admissible set

$$\mathcal{N}_Y = \{N \leq Y : \ (N, mm') = 1, \ v_2(N) \leq 3\},$$

on which the main term of that script is exact, for $Y = 2^{21}$ and $Y = 2^{20}$, and the estimate

$$\|G\|_{\mathrm{est}}^2 = \frac{\mathrm{vol}(\Gamma_0(L)\backslash\mathbb{H})}{16\pi^2} \cdot \frac{1}{\#\mathcal{N}_Y} \sum_{N\in\mathcal{N}_Y} \frac{|a_G(N)|^2}{N}$$

suggested by the Rankin–Selberg asymptotic of Remark 2.1. Replacing $1/Y$ by $1/\#\mathcal{N}_Y$ assumes that the mean square of $|a_G(N)|^2/N$ over the excluded $N$ (those with $(N, mm') > 1$ or $v_2(N) \geq 4$) is the same as over $\mathcal{N}_Y$; this is immaterial when $mm'$ has only large prime factors, but introduces a bias of unknown sign for $(1,5)$, $(9,1)$, $(3,5)$ and $(1,15015)$, where 25%, 38%, 50% and 64% of the $N$ are excluded. Table 1 shows the result at $Y = 2^{21}$, and Figure 1 the dependence on $Y$ and on $mm'$.

| $(m,m')$ | $mm'$ | $\#\mathcal{N}_Y$ | $S_2(Y)/S_2(Y/2)$ | $\|G\|_{\mathrm{est}}^2$ | $\|G\|_{\mathrm{est}}^2/(mm')^{1/2}$ | $\|G\|_{\mathrm{est}}^2/(\log mm')^2$ |
|---|---|---|---|---|---|---|
| $(1,5)$ | 5 | 1572864 | 4.001 | 0.74 | 0.33 | 0.29 |
| $(9,1)$ | 9 | 1310720 | 4.001 | 1.47 | 0.49 | 0.30 |
| $(3,5)$ | 15 | 1048576 | 4.001 | 3.44 | 0.89 | 0.47 |
| $(1,101)$ | 101 | 1946614 | 4.000 | 7.95 | 0.79 | 0.373 |
| $(13,29)$ | 377 | 1752262 | 4.001 | 13.09 | 0.67 | 0.372 |
| $(1,1009)$ | 1009 | 1964131 | 3.976 | 17.96 | 0.57 | 0.375 |
| $(61,67)$ | 4087 | 1904986 | 3.961 | 24.47 | 0.38 | 0.354 |
| $(101,103)$ | 10403 | 1927715 | 3.948 | 29.81 | 0.29 | 0.348 |
| $(1,15015)$ | 15015 | 754220 | 4.359 | 36.62 | 0.30 | 0.396 |
| $(1009,1013)$ | 1022117 | 1962192 | 4.559 | 46.87 | 0.046 | 0.244 |

TABLE 1. Mean square of the cuspidal error over the admissible $N \leq Y = 2^{21}$ (`results/mean_square_check_x21.txt`). The ratio $S_2(Y)/S_2(Y/2) \approx 4$ is the growth $Y^2$ of Theorem 1.2. For the first three pairs and for $(1,15015)$ a large proportion of the $N$ (25% to 64%) is excluded by the coprimality condition, and for the last pair $Y < L$, so that neither (1) in its $Y^2/L$ form nor the Rankin–Selberg asymptotic applies; these rows are shown for completeness. The exact value for $(1,5)$ is 0.5785 (Section 5.2).

5.2. **An exact check at the level 20.** The space $S_2(\Gamma_0(20))$ is one-dimensional, spanned by the newform $f_{20} = \eta(2z)^2\eta(10z)^2$ of the elliptic curve $20a1$: $y^2 = x^3 + x^2 + 4x + 4$, which is $X_0(20)$ itself (modular degree 1, Manin constant 1). Hence $G_{1,5} = c\, f_{20}$ with $c = a_G(1) = T(1,5;1) - M(1,5;1) = 4 - \frac{4}{3} = \frac{8}{3}$ (the Siegel product at $n = 1$ is $\frac{\pi^2}{5} \cdot \frac{25}{3\pi^2} \cdot \frac{4}{5}$), and $\|G_{1,5}\|^2 = \frac{64}{9}\|f_{20}\|^2$. The script `norm_level20.py` computes $\|f_{20}\|^2$ in two independent ways: from the Rankin–Selberg asymptotic $\sum_{n\leq y} a_f(n)^2/n \sim 16\pi^2\|f\|^2 y/\,\mathrm{vol}$, $\mathrm{vol}(\Gamma_0(20)\backslash\mathbb{H}) = 12\pi$, with the coefficients of the eta product up to $2^{20}$, which gives 0.08128, 0.08128, 0.08134 at $y = 2^{16}, 2^{18}, 2^{20}$; and from the period lattice $\Lambda$ of $20a1$, since $\varphi^*\omega = 2\pi i f(z)\mathrm{d}z$ gives $\|f\|^2 = \mathrm{covol}(\Lambda)/(4\pi^2)$ for a modular parametrisation of degree 1. The periods $\omega_1 = 2.824375\ldots$ (real) and $\omega_2 = 2.274165\ldots i$ of $y^2 = (x+1)(x^2+4)$ are computed by quadrature, the lattice $\mathbb{Z}\omega_1 + \mathbb{Z}(\omega_1+\omega_2)/2$ is identified as the period lattice by recomputing $g_2 = -44/3$ and $|g_3| = 296/27$ from it through $E_4$ and $E_6$, and its covolume $3.2115478\ldots$ gives $\|f_{20}\|^2 = 0.0813495\ldots$. The two values agree to four decimals. The script also checks that $T(1,5;N) - M(1,5;N)$ as computed by the pipeline of Part I equals $\frac{8}{3}a_f(N)$ on every admissible $N \leq 2^{20}$, to $10^{-9}$. This validates the Siegel main term, the FFT computation, the Rankin–Selberg normalisation and the volume used in the estimate $\|G\|_{\mathrm{est}}^2$. The exact value $\|G_{1,5}\|^2 = 0.5785$ is below the estimate 0.74 of Table 1, as expected: the excluded $N$ are either multiples of 5, where $|a_f(5n)| = |a_f(n)|$, or divisible by 16, where $a_f = 0$, so the

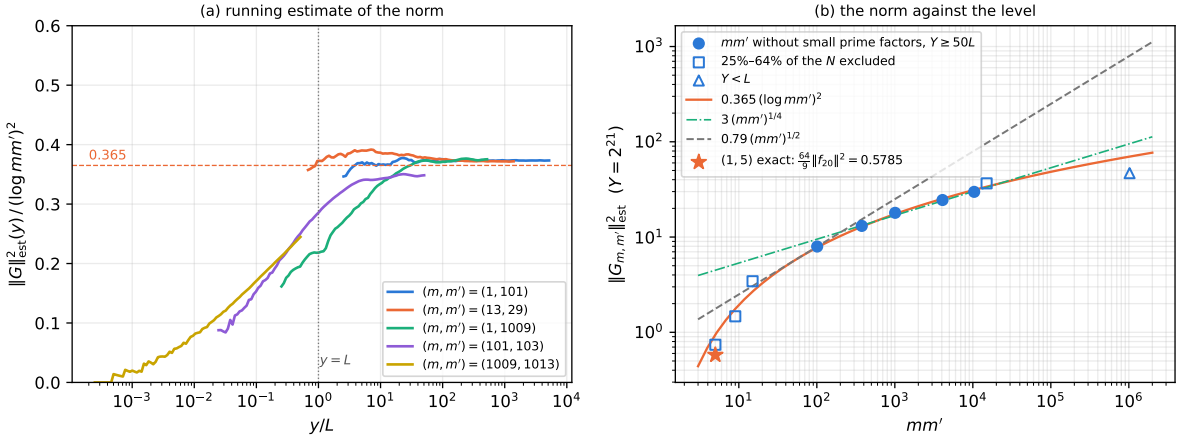



FIGURE 1. (a) The running estimate $\|G\|^2_{\mathrm{est}}(y)$, computed over the admissible $N \le y$, divided by $(\log mm')^2$, against $y/L$ on a logarithmic scale, for five pairs $(m, m')$; the plateau for $y \gg L$ is the growth $Y^2$ of Theorem 1.2 and the Rankin–Selberg asymptotic of Remark 2.1, and its height is the constant 0.365 of Section 5.3 (dashed). (b) $\|G_{m,m'}\|^2_{\mathrm{est}}$ at $Y = 2^{21}$ against $mm'$, both axes logarithmic; filled circles: the five pairs with $mm'$ free of small prime factors and $Y \ge 50L$; open squares: pairs with 25%–64% of the $N$ excluded; triangle: $(1009, 1013)$, for which $Y < L$; star: the exact value at $(1, 5)$. The lines are $0.365(\log mm')^2$, the fit of Section 5.3, $3(mm')^{1/4}$, and $0.79(mm')^{1/2}$ (the growth allowed by Waibel's general bound, through the point $(1, 101)$). Script `plot_mean_square.py`; floating point.

admissible mean is biased upwards. With the main term made exact for every $N$ (Section 6.4), the estimate over all $N \le 2^{21}$ is 0.5785, in agreement with the exact value to four decimals.

5.3. **The empirical law.** The last column of Table 1 shows $\|G\|^2_{\mathrm{est}}/(\log mm')^2$. For the five pairs with $10^2 \le mm' \le 1.1 \cdot 10^4$, $Y \ge 50L$ and $mm'$ free of small prime factors, it lies between 0.348 and 0.375, with mean 0.365, while the ratio to $(mm')^{1/2}$ falls from 0.79 to 0.29 and the ratio to $(mm')^{1/4}$ rises from 2.5 to 3.2 over the same range. Over a range of a factor 100 in $mm'$ the fits $(\log mm')^2$ and $(mm')^{1/4}$ are not separated by the data alone, but Theorem 1.1(i) excludes any power, and the heuristic of Part I, Remark 2.13, that $\|G\|^2$ is a sum over the cusps of $\sum_{n \ll mm'} |b(n)|^2/n$ with $|b(n)| \approx r(\alpha)r(\beta)$ on $O(1)$ pairs, predicts $(\log mm')^2$ with a constant of the order of $\frac{1}{4\pi} \cdot 2 \cdot$ (number of cusp classes with a large contribution) $\approx 0.3$–$0.5$. We therefore record

$$\|G_{m,m'}\|^2 \approx 0.37\,(\log mm')^2 \qquad (10^2 \le mm' \le 1.6 \cdot 10^4),$$

and, for the root mean square of $(T - M)/\sqrt{N}$ over $N \le Y$, $Y \gg L$,

$$\mathrm{rms}^2 \approx \frac{16\pi^2 \|G\|^2}{\mathrm{vol}(\Gamma_0(L)\backslash\mathbb{H})} \approx \frac{44\,(\log mm')^2}{mm' \prod_{p \mid 2mm'}(1 + 1/p)},$$

which reproduces the column "rms" of Table 3 of Part I (0.27 for $(101, 103)$, 0.67 for $(1, 1009)$). This corrects the remark in Section 5.3 of the first two versions of Part I that the root mean square "decreases like $(mm')^{-1/4}$" and that Waibel's bound "appears to be sharp": the decrease is like $(mm')^{-1/2}$ up to a logarithm, as Theorem 1.2 with Theorem 1.1(i) shows; version 3 of Part I states the corrected law. Section 6 shows what the "law" is: up to a few percent,

the norm is the sum of the integrals of $|G|^2$ over the regions above the cusps, which are $\frac{1}{\pi}((\log m)^2+(\log m')^2)+\frac{5}{\pi^2}\log m\log m'+\cdots$ with terms for the other divisors of $mm'$; the constant 0.37 is the average of this over the sample, and the estimates of Table 1 are themselves biased by the restriction to the admissible $N$ (Table 3 has the unbiased values).

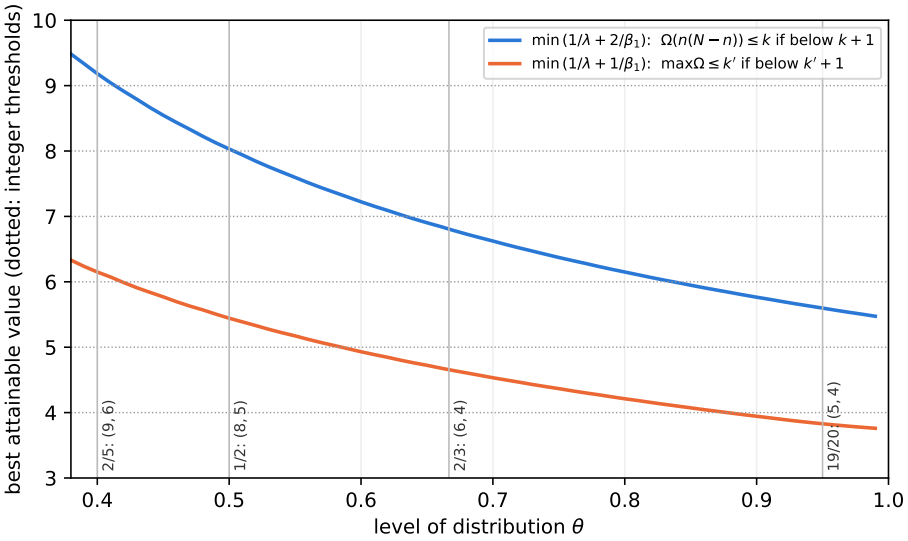



FIGURE 2. The smallest values of $1/\lambda+2/\beta_1$ and of $1/\lambda+1/\beta_1$ attainable with $L-\lambda R>0$ in the sieve of Part I, as functions of the level of distribution $\theta$ (floating-point model of `modular_sieve_scan.py`, minimised over $\alpha$, $\beta_1$ with $\lambda=L/R$; script `plot_sieve_thresholds.py`). The bound $\Omega(n(N-n))\le k$, resp. $\max\Omega\le k'$, holds when the blue, resp. orange, curve is below the dotted line $k+1$, resp. $k'+1$. The vertical lines mark $\theta=2/5$ (Waibel's bound alone) and $1/2$ (Part I, every $N$), $2/3$ and $19/20$ (almost all $N$); the parameter sets at these four levels are certified.

5.4. **The supremum over $N$.** Conjecture 4.1 says that the supremum of $|a_G(N)|/\sqrt{N}$ over $N$ in a range is, up to $N^{\varepsilon}L^{\varepsilon}$, of the size of the root mean square $\approx 2.7(\log mm')(mm')^{-1/2}$ of Section 5.3. Table 2, extracted from the scans of Part I (`results/sup_error_scan_x21.txt`, `_x23.txt` there), gives the ratio of the supremum over all admissible $N\in(X/2,X]$ to the root mean square, for $X=2^{21}$ (about $10^6$ values of $N$ per pair) and $X=2^{23}$ (about $4\cdot10^6$). For the pairs with $L\le 6\cdot10^4$ the ratio lies between 5.4 and 9.8 and shows no trend in $mm'$; its mild growth from $X=2^{21}$ to $2^{23}$ is the growth of the extreme value of a fixed heavy-tailed distribution with the sample size (the suprema are attained at $N$ with 16–192 divisors, in line with the factor $d(N)$ of Deligne's bound). The pair $(9,1)$ is special: $S_2(\Gamma_0(36))$ is spanned by the newform of the curve $36a1$, $y^2=x^3+1$, which has complex multiplication, so that $a_f(p)=0$ for $p\equiv 2 \pmod 3$ and the distribution of $a_f(N)$ is far from the Gaussian bulk of Figure 3 of Part I. The pair $(1009,1013)$ is in the range $X<2L$, where neither the mean square nor the supremum has reached its asymptotic regime.

| $(m,m')$ | $mm'$ | rms | sup, $X=2^{21}$ | sup /rms | sup, $X=2^{23}$ | sup /rms |
|---|---|---|---|---|---|---|
| $(1,5)$ | 5 | 1.76 | 14.7 | 8.4 | 17.3 | 9.8 |
| $(3,5)$ | 15 | 1.90 | 10.5 | 5.5 | 12.2 | 6.4 |
| $(1,101)$ | 101 | 1.40 | 11.8 | 8.4 | 12.1 | 8.6 |
| $(13,29)$ | 377 | 0.89 | 5.9 | 6.6 | 6.8 | 7.7 |
| $(1,1009)$ | 1009 | 0.67 | 4.4 | 6.5 | 5.3 | 8.0 |
| $(61,67)$ | 4087 | 0.38 | 2.5 | 6.5 | 2.4 | 6.3 |
| $(101,103)$ | 10403 | 0.27 | 1.9 | 7.3 | 1.8 | 6.7 |
| $(1,15015)$ | 15015 | 0.18 | 1.0 | 5.7 | – | – |
| $(9,1)$ | 9 | 1.76 | 28.7 | 16.3 | 33.8 | 19.2 |
| $(1009,1013)$ | 1022117 | 0.036 | 0.61 | 17 | 0.55 | 14 |

TABLE 2. The supremum of $|T(m,m';N)-M(m,m';N)|/\sqrt{N}$ over the admissible $N\in(X/2,X]$ of Part I, Section 5, and its ratio to the root mean square over the same $N$ (the rms is the same to two decimals for both $X$). The last two pairs are the degenerate cases discussed in the text.

## 6. A lower bound for the norm: the regions above the cusps

Throughout this section $m,m'$ are coprime odd positive integers, $L=4mm'$, $\chi=\chi_{-4}$, and for $\gamma=\left(\begin{smallmatrix}a&b\\c&d\end{smallmatrix}\right)\in\mathrm{GL}_2^+(\mathbb{Q})$ and $k\in\{1,2\}$ we write $(f|_k\gamma)(z)=(\det\gamma)^{k/2}(cz+d)^{-k}f(\gamma z)$; thus scalar matrices act trivially, $f(mz)=m^{-k/2}(f|_kB_m)(z)$ with $B_m=\left(\begin{smallmatrix}m&0\\0&1\end{smallmatrix}\right)$, and $(fg)|_2\gamma=(f|_1\gamma)(g|_1\gamma)$. We use $\theta^2\in M_1(\Gamma_0(4),\chi)$, i.e. $\theta^2|_1\gamma=\chi(d)\theta^2$ for $\gamma\in\Gamma_0(4)$, and $\theta(-1/(4z))=\sqrt{-2iz}\,\theta(z)$, which gives $\theta^2|_1\left(\begin{smallmatrix}0&-1\\4&0\end{smallmatrix}\right)=-i\theta^2$.

6.1. **The Atkin–Lehner cusps.** For $Q\mid L$ with $(Q,L/Q)=1$, an Atkin–Lehner matrix is $W_Q=\left(\begin{smallmatrix}Qa&b\\Lc&Qd\end{smallmatrix}\right)$ with $a,b,c,d\in\mathbb{Z}$ and $\det W_Q=Q$ [1]. It normalises $\Gamma_0(L)$; hence for $f\in M_2(\Gamma_0(L))$ the form $f|_2W_Q$ lies in $M_2(\Gamma_0(L))$ and does not depend on the choice of $W_Q$ (two choices differ by an element of $\Gamma_0(L)$ on either side), $f\mapsto f|_2W_Q$ is an isometry for the Petersson product which maps cusp forms to cusp forms, and it therefore preserves the orthogonal decomposition of $M_2(\Gamma_0(L))$ into the Eisenstein and the cuspidal subspace.

**Lemma 6.1.** *Let $Q\in\{m,4m\}$. Then $\Theta_{m,m'}|_2W_Q=\varepsilon_Q\,\Theta_{1,mm'}$ with $|\varepsilon_Q|=1$, and consequently $E_{m,m'}|_2W_Q=\varepsilon_QE_{1,mm'}$ and $G_{m,m'}|_2W_Q=\varepsilon_QG_{1,mm'}$. The same holds with $m$ and $m'$ interchanged.*

*Proof.* Let first $Q=m$ and $W_m=\left(\begin{smallmatrix}ma&b\\Lc&md\end{smallmatrix}\right)$, $mad-4m'bc=1$. Then $B_mW_m=m\left(\begin{smallmatrix}ma&b\\4m'c&d\end{smallmatrix}\right)$ with the second factor in $\Gamma_0(4)$, so $\theta^2(m\cdot)|_1W_m=m^{-1/2}\theta^2|_1(B_mW_m)=m^{-1/2}\chi(d)\theta^2$; and $W_m=\left(\begin{smallmatrix}a&b\\4m'c&md\end{smallmatrix}\right)B_m$ with the first factor in $\Gamma_0(4m')$, so $\theta^2(m'\cdot)|_1W_m=\chi(md)\,\theta^2(m'\cdot)|_1B_m=\chi(md)\,m^{1/2}\theta^2(mm'\cdot)$. Multiplying, $\Theta_{m,m'}|_2W_m=\chi(m)\,\theta^2(z)\theta^2(mm'z)=\chi(m)\Theta_{1,mm'}$.

Let now $Q=4m$ and $W_{4m}=\left(\begin{smallmatrix}4ma&b\\Lc&4md\end{smallmatrix}\right)$, $4mad-m'bc=1$. Then $B_mW_{4m}=m\left(\begin{smallmatrix}4ma&b\\4m'c&4d\end{smallmatrix}\right)$, and the second factor is an Atkin–Lehner matrix $W_4$ for $\Gamma_0(4)$; every such matrix is $\left(\begin{smallmatrix}0&-1\\4&0\end{smallmatrix}\right)$ times an element of $\Gamma_0(4)$, so $\theta^2|_1W_4=\pm i\theta^2$ and $\theta^2(m\cdot)|_1W_{4m}=\pm i\,m^{-1/2}\theta^2$. Also $W_{4m}=W_4'B_m$ with $W_4'=\left(\begin{smallmatrix}4a&b\\4m'c&4md\end{smallmatrix}\right)$ an Atkin–Lehner matrix for $\Gamma_0(4m')$, and $B_{m'}W_4'=\left(\begin{smallmatrix}4a&m'b\\4c&4md\end{smallmatrix}\right)B_{m'}$ with an Atkin–Lehner matrix for $\Gamma_0(4)$ in front, so $\theta^2(m'\cdot)|_1W_4'=m'^{-1/2}(\pm i\theta^2)|_1B_{m'}=\pm i\,\theta^2(m'\cdot)$ and $\theta^2(m'\cdot)|_1W_{4m}=\pm i\,m^{1/2}\theta^2(mm'\cdot)$. Multiplying, $\Theta_{m,m'}|_2W_{4m}=\pm\Theta_{1,mm'}$. The assertions on $E$ and $G$ follow because $|_2W_Q$ preserves the orthogonal decomposition and $\Theta_{1,mm'}=E_{1,mm'}+G_{1,mm'}$ is that decomposition for $\Theta_{1,mm'}$ (Siegel's theorem applied to $Q_{1,mm'}$). □

We now use the formula of Part I, Lemma 2.8 (Waibel's [7, (22)]): with $\mu=[\mathrm{SL}_2(\mathbb{Z}):\Gamma_0(L)]$, $\tilde\mu=[\mathrm{SL}_2(\mathbb{Z}):\Gamma(L)]$, $\tilde\mu/\mu=L\varphi(L)$, $\mathcal{F}$ the standard fundamental domain of $\mathrm{SL}_2(\mathbb{Z})$, $T=\{\left(\begin{smallmatrix}1&t\\0&1\end{smallmatrix}\right):$

$0 \le t \le L-1\}$ and $\varrho_{a,c} = \left(\begin{smallmatrix} a & * \\ c & * \end{smallmatrix}\right) \in \mathrm{SL}_2(\mathbb{Z})$,

$$\|G\|^2 = \frac{\mu}{\tilde{\mu}} \sum_{c \ (L)} \sum_{\substack{a \ (L) \\ (a,c,L)=1}} I(\varrho_{a,c}), \qquad I(\varrho) = \int_{\bigcup_{t\in T} t\mathcal{F}} \left|(G|_2\varrho)(z)\right|^2 \mathrm{d}x\,\mathrm{d}y, \tag{10}$$

for $G \in S_2(\Gamma_0(L))$, where $(a,c)$ runs over coprime representatives of the residue classes. Every term is non-negative, and $\bigcup_{t\in T} t\mathcal{F} \supseteq \{-\frac{1}{2} \le x \le L-\frac{1}{2},\ y \ge 1\}$.

**Lemma 6.2.** *Let $Q \in \{m, 4m, m', 4m'\}$ and $a(n) = a_{G_{1,mm'}}(n)$. The terms of (10) for $G = G_{m,m'}$ with $(c, L) = L/Q$ contribute in total at least*

$$\frac{1}{4\pi} \sum_{n\ge 1} \frac{|a(n)|^2}{n}\, e^{-4\pi n/Q}.$$

*Proof.* Let $(c, L) = L/Q$, $c = (L/Q)c_1$ with $(c_1, Q) = 1$, and $(a, c) = 1$; then $(a, L/Q) = 1$, so $(aQ, (L/Q)c_1) = 1$ and there are $b, d \in \mathbb{Z}$ with $aQd - b(L/Q)c_1 = 1$. The matrix $\varrho = \left(\begin{smallmatrix} a & b \\ (L/Q)c_1 & Qd \end{smallmatrix}\right)$ is in $\mathrm{SL}_2(\mathbb{Z})$, has the prescribed first column, and $\varrho B_Q = \left(\begin{smallmatrix} Qa & b \\ Lc_1 & Qd \end{smallmatrix}\right)$ is an Atkin–Lehner matrix $W_Q$. Hence, by Lemma 6.1 and $B_Q^{-1} = Q^{-1}\left(\begin{smallmatrix} 1 & 0 \\ 0 & Q \end{smallmatrix}\right)$,

$$G_{m,m'}|_2\varrho = (G_{m,m'}|_2 W_Q)|_2 B_Q^{-1} = \varepsilon_Q\, G_{1,mm'}|_2 \left(\begin{smallmatrix} 1 & 0 \\ 0 & Q \end{smallmatrix}\right) = \frac{\varepsilon_Q}{Q}\, G_{1,mm'}\Big(\frac{z}{Q}\Big) = \frac{\varepsilon_Q}{Q} \sum_{n\ge 1} a(n)\, e\Big(\frac{nz}{Q}\Big).$$

This function has period $Q$ in $x$, and $Q \mid L$, so by Parseval over $-\frac{1}{2} \le x \le L - \frac{1}{2}$,

$$I(\varrho) \ \ge\ \int_1^\infty \int_{-1/2}^{L-1/2} |G|_2\varrho|^2 \,\mathrm{d}x\,\mathrm{d}y = \frac{L}{Q^2} \sum_{n\ge 1} |a(n)|^2 \int_1^\infty e^{-4\pi n y/Q}\,\mathrm{d}y = \frac{L}{4\pi Q} \sum_{n\ge 1} \frac{|a(n)|^2}{n} e^{-4\pi n/Q}.$$

The number of pairs $(a, c)$ modulo $L$ with $(c, L) = L/Q$ and $(a, c, L) = 1$, i.e. $(a, L/Q) = 1$, is $\varphi(Q)\cdot\varphi(L/Q)\,Q = \varphi(L)Q$, and $\mu/\tilde{\mu} = 1/(L\varphi(L))$. Multiplying gives the assertion. □

*Proof of Theorem 1.5(i).* Since $mm' > 1$ and $(m, m') = 1$ we have $m \ne m'$, so the four values $L/Q$, $Q \in \{m, 4m, m', 4m'\}$, are distinct and the four sets of pairs $(a, c)$ in Lemma 6.2 are disjoint. As all terms of (10) are non-negative, (i) follows by adding the four lower bounds. □

6.2. **The sum $\sum r(n)^2/n$.**

**Lemma 6.3.** *For $Y \ge 1$,*

$$S(Y) := \sum_{n\ge 1} \frac{r(n)^2}{n}\, e^{-n/Y} = 2(\log Y)^2 + c_1 \log Y + c_2 + O(Y^{-1/2}),$$

$$c_1 = 4\gamma + 8\frac{L'}{L}(1,\chi) - 8\frac{\zeta'}{\zeta}(2) + \frac{4}{3}\log 2 = 9.758\ldots,$$

*with an absolute constant $c_2$ ($c_2 = 6.12\ldots$ numerically); in particular $S(Y) \ge 2(\log Y)^2 - C$ for $Y \ge 1$. Moreover $\sum_{n\ge 1} r(n)^2 e^{-n/X} = 4X\log X + O(X)$, so that $\sum_{n\le X} r(n)^2 \ll X\log 2X$, for $X \ge 1$.*

*Proof.* Since $r(n)/4 = \sum_{d\mid n} \chi(d)$ is multiplicative with $r(p^k)/4 = k+1$, $\frac{1}{2}(1+(-1)^k)$, $1$ for $p \equiv 1, 3 \pmod 4$ and $p = 2$, the series $\sum_k (r(p^k)/4)^2 x^k$ equals $(1+x)(1-x)^{-3}$, $(1-x^2)^{-1}$, $(1-x)^{-1}$ respectively, and

$$D(s) = \sum_{n\ge 1} \frac{r(n)^2}{n^s} = \frac{16\,\zeta(s)^2 L(s,\chi)^2}{(1+2^{-s})\,\zeta(2s)} \qquad (\mathrm{Re}\, s > 1).$$

By Mellin inversion, $S(Y) = \frac{1}{2\pi i}\int_{(1)} \Gamma(w) Y^w D(1+w)\,\mathrm{d}w$. We move the contour to $\operatorname{Re} w = -\frac12$: there $\zeta(2+2w) \neq 0$ and $1/\zeta(2+2w) \ll \log(|w|+2)$, $\zeta(1+w)$ and $L(1+w,\chi)$ are $\ll (|w|+2)^{1/4}$, and $\Gamma(w)$ decays exponentially, so the integral on the new line is $O(Y^{-1/2})$. The only pole crossed is $w=0$, of order three. With $\Gamma(w) = w^{-1}(1-\gamma w + O(w^2))$, $Y^w = 1 + w\log Y + \frac12 w^2(\log Y)^2 + O(w^3)$ and $D(1+w) = 4w^{-2}(1 + a_1 w + O(w^2))$, where $16 \cdot \frac23 \cdot L(1,\chi)^2/\zeta(2) = 16 \cdot \frac23 \cdot \frac{\pi^2}{16} \cdot \frac{6}{\pi^2} = 4$ and $a_1 = 2\gamma + 2\frac{L'}{L}(1,\chi) - 2\frac{\zeta'}{\zeta}(2) + \frac13 \log 2$, the residue is $2(\log Y)^2 + 4(a_1 - \gamma)\log Y + c_2$. The numerical value uses $L'(1,\chi)/L(1,\chi) = \gamma + 2\log 2 + 3\log\pi - 4\log\Gamma(\frac14) = 0.2456\ldots$ and $\zeta'(2)/\zeta(2) = -0.5700\ldots$. Since $c_1 > 0$, $S(Y) \geq 2(\log Y)^2 + c_2 - O(1)$ for $Y \geq 1$. The last assertion is the same argument for $\frac{1}{2\pi i}\int \Gamma(w) X^w D(w)\mathrm{d}w$ with the contour moved to $\operatorname{Re} w = \frac34$, where only the double pole at $w=1$ is crossed, and $\sum_{n\leq X} r(n)^2 \leq e\sum_n r(n)^2 e^{-n/X}$. □

### 6.3. **The main term for small $n$.**

**Lemma 6.4.** *Let $Q = Q_{m,m'}$ and $n \geq 1$. Then $\beta_2(n) \leq 2$; $\beta_p(n) \leq \sigma_{-1}(p^{v_p(n)})$ for $p \nmid 2mm'$; and $\beta_p(n) \leq 7\,(\min(v_p(n), v_p(mm')) + 1)$ for $p \mid mm'$, while $\beta_p(n) \leq 1 + 1/p$ for $p \mid mm'$, $p \nmid n$. Consequently, for $Y \geq 1$,*

$$\sum_{n\geq 1} \frac{r(n)\,M(m,m';n)}{n}\, e^{-n/Y} \;\ll\; \frac{Y}{mm'}\,\log(2Y)\,(\log\log 3mm')^C$$

*with an absolute constant $C$.*

*Proof.* For a prime $p$, $a \geq 1$ and $t \in \mathbb{Z}$ let $N_2(t;p^a) = \#\{(x,y) \bmod p^a :\ x^2+y^2 \equiv t\}$. We claim

$$N_2(t;p^a) \leq 2\,(\min(v_p(t), a) + 1)\,p^a \quad (p \text{ odd}), \qquad N_2(t;2^a) \leq 2^{a+1}. \tag{11}$$

For odd $p$, the solutions with $(x,y) \not\equiv (0,0) \pmod p$ are non-singular and lift from $p$ to $p^a$ with multiplicity $p^{a-1}$; modulo $p$ there are $p - \chi(p) \leq 2p$ of them if $p \nmid t$ and $2(p-1)$ or $0$ if $p \mid t$. The solutions with $x \equiv y \equiv 0 \pmod p$ exist only if $p^2 \mid t$ and are $p^2 N_2(t/p^2; p^{a-2})$ in number for $a \geq 2$. Hence $N_2(t;p^a) \leq 2p^a + p^2 N_2(t/p^2;p^{a-2})$, and (11) follows by induction on $a$ ($N_2(t;p) \leq 2p$, $N_2(t;1) = 1$). For $p = 2$ and $a \geq 3$: if $t \equiv 1 \pmod 4$, exactly one of $x, y$ is odd, and since the odd squares modulo $2^a$ are the residues $\equiv 1 \pmod 8$, each taken four times, while $y^2 \equiv 0$ or $4 \pmod 8$ according as $4 \mid y$ or not, there are $2 \cdot 4 \cdot 2^{a-2} = 2^{a+1}$ solutions; if $t \equiv 2 \pmod 8$ both are odd and there are $4 \cdot 4 \cdot 2^{a-3} = 2^{a+1}$ solutions; if $t \equiv 3 \pmod 4$ or $t \equiv 6 \pmod 8$ there are none; and if $4 \mid t$ both are even and $N_2(t;2^a) = 4N_2(t/4;2^{a-2})$, so (11) follows by induction from $N_2(t;2^a) \leq 2^{a+1}$ for $a \leq 2$, which one checks by hand.

Now $\beta_2(n) = \lim_a 2^{-3a}\sum_{(z,w) \bmod 2^a} N_2\big((n - m'(z^2+w^2))\bar m; 2^a\big) \leq \lim_a 2^{-3a} \cdot 2^{2a} \cdot 2^{a+1} = 2$, where $\bar m$ is an inverse of $m$ modulo $2^a$. For $p \nmid 2mm'$ the formula $\beta_p(n) = (1-p^{-2})\sigma_{-1}(p^{v_p(n)})$ is (6) of Part I. Let $p \mid mm'$, say $p^e \parallel m$ with $e \geq 1$, $p \nmid m'$, and $p^k \parallel n$. By (11), with $X = x^2+y^2$,

$$\beta_p(n) = \lim_a p^{-3a} \sum_{(x,y) \bmod p^a} N_2\big((n - mX)\bar m'; p^a\big) \leq 2\Big(1 + \sum_{i\geq 1} \operatorname{Prob}_{(x,y)}\big(p^i \mid n - mX\big)\Big),$$

the probability referring to $(x,y)$ uniform modulo $p^a$ with $a \geq i$. For $i \leq \min(k,e)$ the probability is 1. If $k < e$ it is 0 for $i > k$. If $k \geq e$ and $i > e$, then $p^i \mid n - mX$ if and only if $X \equiv n_1 \pmod{p^{i-e}}$ with $n_1 = (n/p^e)(m/p^e)^{-1}$, which has probability $N_2(n_1;p^{i-e})p^{-2(i-e)} \leq 2(i-e+1)p^{-(i-e)}$ by (11), and $\sum_{j\geq 1} 2(j+1)p^{-j} \leq \frac52$ for $p \geq 3$. Hence $\beta_p(n) \leq 2(\min(k,e) + \frac72) \leq 7(\min(k,e)+1)$; for $k = 0$ the exact value $1 - \chi(p)/p$ is (7) of Part I.

For the sum, write $M(m,m';n) = \pi^2 n\,\beta(n)/(mm')$ with $\beta(n) = \prod_p \beta_p(n)$, so that the sum is $\frac{\pi^2}{mm'}\sum_n r(n)\beta(n)e^{-n/Y}$, and $\beta(n) \leq 2\,\sigma_{-1}(n)\prod_{p\mid mm'}(1+\frac1p)\prod_{p\mid(n,mm')} 7(v_p(n)+1)$. Write $n = gk$ with $g$ composed of the primes of $mm'$ and $(k, mm') = 1$; then $r(n) \leq 4d(g)d(k)$,

$\sigma_{-1}(n)\le 2^{\omega(g)}\sigma_{-1}(k)$, and $\sum_{k\le x} d(k)\sigma_{-1}(k)\ll x\log 2x$ (write $\sigma_{-1}(k)=\sum_{u|k}1/u$ and sum over $u$ first). Therefore

$$\sum_n r(n)\beta(n)e^{-n/Y}\ \ll\ \prod_{p|mm'}\Big(1+\frac{1}{p}\Big)\sum_g \frac{7^{\omega(g)}\,2^{\omega(g)}\,d(g)^2}{g}\,Y\log(2Y)\ \ll\ Y\log(2Y)\prod_{p|mm'}\Big(1+\frac{C}{p}\Big),$$

since $\sum_{j\ge1}14(j+1)^2p^{-j}\le C/p$ for $p\ge3$, and $\prod_{p|mm'}(1+C/p)\le\exp(C\sum_{p|mm'}1/p)\ll(\log\log 3mm')^C$. □

*Proof of Theorem 1.5(ii).* Fix $Q\in\{m,4m,m',4m'\}$ with $Q\ge4\pi$ and put $Y=Q/(4\pi)\ge1$; note $Y\le\max(m,m')/\pi\le mm'/\pi$. Keep only $n<mm'$ in the inner sum of (i), where $a(n)=r(n)-M(1,mm';n)$ and $M\ge0$, so that $|a(n)|^2\ge r(n)^2-2r(n)M(1,mm';n)$:

$$\sum_{n\ge1}\frac{|a(n)|^2}{n}e^{-n/Y}\ \ge\ S(Y)-\sum_{n\ge mm'}\frac{r(n)^2}{n}e^{-n/Y}-2\sum_{n\ge1}\frac{r(n)M(1,mm';n)}{n}e^{-n/Y}.$$

By Lemma 6.3, $S(Y)\ge2(\log Y)^2-C$, and the tail is at most

$$\sum_{j\ge0}e^{-\pi2^j}\,\frac{1}{2^jmm'}\sum_{n<2^{j+1}mm'}r(n)^2\ \ll\ \sum_{j\ge0}e^{-\pi2^j}\log(2^{j+2}mm')\ \ll\ \log(2mm').$$

By Lemma 6.4 with $(m,m')$ replaced by $(1,mm')$, the last sum is $\ll(Y/mm')\log(2Y)(\log\log 3mm')^C\le\log(2mm')(\log\log 3mm')^C$, because $Y/mm'\le1/\min(m,m')\le1$. Summing over the four $Q$ and using

$$2\big(\log(m/4\pi)\big)^2+2\big(\log(m/\pi)\big)^2\ \ge\ 4(\log m)^2-4\log(4\pi^2)\log m$$

(and the same for $m'$), while the terms with $Q<4\pi$, if any, are dropped and the corresponding $(\log m)^2\le(\log 12)^2$ is absorbed in the constant, (i) gives

$$\|G_{m,m'}\|^2\ \ge\ \frac{1}{\pi}\Big((\log m)^2+(\log m')^2\Big)-\frac{\log(4\pi^2)}{\pi}\log(mm')-C\log(2mm')(\log\log 3mm')^C,$$

which is (ii); the second inequality in (ii) is $(\log m)^2+(\log m')^2\ge\frac12(\log mm')^2$. □

6.4. **All cusps, numerically.** For square-free $m,m'$ the computation of Lemma 6.2 can be carried out at every cusp. The cusps of $\Gamma_0(L)$ are then in bijection with the divisors $d$ of $L$ (one cusp of denominator $d$ for each $d$, since $(d,L/d)\le2$), and we write $d=d_2\delta$ with $d_2\mid4$, $\delta\mid mm'$, $\delta_m=(\delta,m)$, $\delta_{m'}=(\delta,m')$, and

$$A=\delta_m\,\frac{m'}{\delta_{m'}},\qquad B=\delta_{m'}\,\frac{m}{\delta_m},\qquad AB=mm',\quad (A,B)=1.$$

The cusps with $d_2\in\{1,4\}$ are the images of $\infty$ under the Atkin–Lehner involutions $W_{L/d}$, their widths are $L/d=4mm'/\delta$ and $mm'/\delta$, and the proof of Lemma 6.1, applied prime by prime, gives $G_{m,m'}|_2W_{L/d}=\varepsilon\,G_{A,B}$ with $|\varepsilon|=1$ (the odd part of $L/d$ moves the primes of $\delta_m$ from the first block to the second and those of $\delta_{m'}$ from the second to the first; the 2-part contributes a sign). The two cusps of Theorem 1.5 with the pair $(1,mm')$ are $\delta\in\{m,m'\}$. At the cusp with $d_2=2$, the function $\theta^2$ is transformed into $\theta_2^2$, $\theta_2(z)=\sum_{x\text{ odd}}e(x^2z/4)$ (from $\theta^2|_1\left(\begin{smallmatrix}1&0\\2&1\end{smallmatrix}\right)=\theta_2^2$, which follows from the transformation formulae of $\theta_3,\theta_4,\theta_2$ under $z\mapsto-1/(4z)$), and $\Theta_{m,m'}$ into $(mm'/\delta)^{-1}\sum_n T^{\mathrm{odd}}(A,B;n)e(\delta nz/(4mm'))$, where $T^{\mathrm{odd}}$ counts the representations of $n$ by $Q_{A,B}$ with all four coordinates odd; the Eisenstein part of this expansion is given by Siegel's theorem for the coset $(1+2\mathbb{Z})^4$ of $2\mathbb{Z}^4$, i.e. by $M^{\mathrm{odd}}(A,B;n)=\frac{\pi^2n}{AB}\cdot\frac12[n\equiv2(A+B)\ (8)]\prod_{p>2}\beta_p(n)$ (for $(A,B)=(1,1)$ this is Jacobi's formula $16\sigma(n/4)$ for the number of representations of $n\equiv4$

(mod 8) as a sum of four odd squares). Integrating $|G|^2$ over the region above each cusp as in Lemma 6.2 gives

$$\|G_{m,m'}\|^2 \;\ge\; \Lambda(m,m') := \sum_{\delta\mid mm'} \Big[\frac{1}{4\pi}\sum_{n\ge1}\frac{|a_{G_{A,B}}(n)|^2}{n}\big(e^{-\pi\delta n/mm'}+e^{-4\pi\delta n/mm'}\big) + \frac{1}{\pi}\sum_{n\ge1}\frac{|T^{\mathrm{odd}}(A,B;n)-M^{\mathrm{odd}}(A,B;n)|^2}{n}\,e^{-\pi\delta n/mm'}\Big], \tag{12}$$

where $A, B$ depend on $\delta$ as above. The regions above all the cusps have total area equal to the sum of the widths, which is $\mu = L\prod_{p\mid L}(1+1/p)$, i.e. the fraction $3/\pi = 0.955$ of $\mathrm{vol}(\Gamma_0(L)\backslash\mathbb{H})$; the rest of the fundamental domain lies below height 1 at every cusp.

The script `cusp_lower_bound.py` evaluates (12). It needs the main terms $M(A,B;n)$ and $M^{\mathrm{odd}}(A,B;n)$ for *every* $n$, not only the admissible ones: the module `local_densities.py` computes the densities at the primes $p \mid AB$ dividing $n$ by the recursion $\beta_p(Q_{A,B};pn') = 2(1-\frac1p)[p\equiv1\ (4)] + \frac1p\beta_p(Q_{A/p,pB};n')$ for $p\mid A$ (the non-singular solutions have $(z,w)\not\equiv0$, the singular ones reduce to the form $Q_{A/p,pB}$), which gives $\beta_p(n) = (1+\frac1p)p^{-k}$ for $p\equiv3$ and $2-(1+\frac1p)p^{-k}$ for $p\equiv1\pmod4$, $k=v_p(n)$; the 2-adic density by sorting the solutions by the parities of $(x,y)$ and $(z,w)$, which gives $\beta_2(n) = \frac14\beta_2(n/4) + \frac12[A\equiv B\ (4)][v_2(n)=2] + [A\not\equiv B\ (4)](1+\frac12[v_2(n)\ge3])$ for $v_2(n)\ge2$; and $\beta_2^{\mathrm{odd}}(n) = \frac12[n\equiv 2(A+B)\ (8)]$. These formulae are checked exactly: at the levels 4 and 12, where there are no cusp forms, $T=M$ and $T^{\mathrm{odd}}=M^{\mathrm{odd}}$ for every $n\le2^{17}$ to $10^{-9}$, and at the level 20, $T-M=\frac83 a_{f_{20}}(n)$ for every $n\le2^{17}$ (Section 5.2 had this only for the admissible $n$), while the mean square of $T^{\mathrm{odd}}-M^{\mathrm{odd}}$ grows like $Y^2$ as for a cusp form. With the exact main term the Rankin–Selberg estimate of Section 5 can be taken over all $n\le Y$; this is the column $\|G\|^2$ of Table 3, and for $(1,5)$ it is 0.5785, the exact value of Section 5.2.

| $(m,m')$ | $mm'$ | $\|G\|^2$ | Tab. 1 | $\Lambda_{\mathrm{AL}}$ | $\Lambda_{\mathrm{odd}}$ | $\Lambda$ | $\Lambda/\|G\|^2$ | Thm 1.5(i) | $\frac1\pi(\log^2 m+\log^2 m')$ |
|---|---|---|---|---|---|---|---|---|---|
| $(1,5)$ | 5 | 0.5785 | 0.74 | 0.496 | 0.046 | 0.543 | 0.938 | 0.496 | 0.82 |
| $(3,5)$ | 15 | 2.058 | 3.44 | 1.799 | 0.136 | 1.936 | 0.940 | 1.061 | 1.21 |
| $(3,7)$ | 21 | 2.481 | 4.02 | 2.186 | 0.167 | 2.352 | 0.948 | 1.183 | 1.59 |
| $(5,7)$ | 35 | 4.109 | 4.82 | 3.555 | 0.314 | 3.869 | 0.942 | 2.154 | 2.03 |
| $(1,101)$ | 101 | 7.429 | 7.95 | 7.147 | 0.100 | 7.247 | 0.976 | 7.147 | 6.78 |
| $(13,29)$ | 377 | 12.423 | 13.09 | 10.795 | 1.033 | 11.828 | 0.952 | 6.639 | 5.70 |
| $(15,77)$ | 1155 | 21.779 | 27.93 | 19.620 | 1.340 | 20.960 | 0.962 | 9.596 | 8.34 |
| $(1,1009)$ | 1009 | 16.937 | 17.96 | 16.572 | 0.106 | 16.678 | 0.985 | 16.572 | 15.23 |
| $(61,67)$ | 4087 | 24.704 | 24.47 | 21.411 | 2.000 | 23.411 | 0.948 | 12.875 | 11.01 |
| $(101,103)$ | 10403 | 30.385 | 29.81 | 26.470 | 2.447 | 28.917 | 0.952 | 15.847 | 13.62 |
| $(105,143)$ | 15015 | 47.140 | 49.50 | 40.622 | 3.186 | 43.807 | 0.929 | 17.079 | 14.73 |
| $(1,15015)^*$ | 15015 | 43.233 | 43.54 | 44.610 | 0.230 | 44.839 | 1.037 | 30.818 | 29.44 |
| $(1009,1013)^*$ | 1022117 | 57.730 | 61.23 | 58.545 | 5.271 | 63.816 | 1.105 | 34.469 | 30.47 |

TABLE 3. The lower bound (12) and its parts (`results/cusp_lower_bound.txt`, `_large.txt`): $\Lambda_{\mathrm{AL}}$ is the sum over the Atkin–Lehner cusps ($d_2\in\{1,4\}$), $\Lambda_{\mathrm{odd}}$ the sum over the cusps with $d_2=2$, $\Lambda$ their sum, "Thm 1.5(i)" the right-hand side of Theorem 1.5(i) (the four cusps with $\delta\in\{m,m'\}$), and the last column its leading term. $\|G\|^2$ is the Rankin–Selberg estimate over all $n\le2^{21}$ with the exact main term ($2^{24}$ for the two pairs marked $^*$), and "Tab. 1" the estimate over the admissible $n$ of Section 5 (for $(3,7)$, $(5,7)$, $(15,77)$, $(105,143)$ computed here in the same way). For the two pairs marked $^*$ the estimate has not converged ($Y/L=280$ and 4; Table 1 has $S_2(Y)/S_2(Y/2)=4.36$ and 4.56 for them), and the lower bound is the better information. Floating point.

Table 3 shows the following.

(a) For the ten pairs with $Y \geq 50L$, the regions above the cusps carry between 93.8% and 98.5% of the norm (for $(105, 143)$, with $Y/L = 35$, 92.9%), about their share 95.5% of the area: the function $y^2|G|^2$ has no excess mass in the remaining 4.5% of the fundamental domain, near the elliptic points and in the "corners" between the cusps. The lower bound (12) is thus nearly an equality, and an asymptotic formula for $\|G_{m,m'}\|^2$ is a question about the lower part of the fundamental domain only.

(b) The four cusps of Theorem 1.5(i) alone give 98% for $m = 1$, where they are all the Atkin–Lehner cusps, and about half for $mm' = pq$; the rest comes from the pair $(m, m')$ itself at the cusps with $\delta = 1$ (widths $4mm'$ and $mm'$, and the cusp of denominator 2): for $(13, 29)$, $4.16 + 1.03$ out of 12.42.

(c) For the cusps with $\delta = 1$, $T(m, m'; n)$, $n \ll mm'$, is a sum of at most $1 + n/mm'$ products $r(\alpha)r(\beta)$ with $m\alpha + m'\beta = n$; replacing $r(\alpha)^2, r(\beta)^2$ by their mean values $4\log\alpha$, $4\log\beta$ (Lemma 6.3) gives for their Atkin–Lehner part

$$\frac{16}{4\pi}\iint_{u,v>0}\frac{\log(m'u)\log(mv)}{u+v}\big(e^{-\pi(u+v)}+e^{-4\pi(u+v)}\big)\mathrm{d}u\,\mathrm{d}v = \frac{5}{\pi^2}\log m\log m' + O(\log mm'),$$

since $\iint e^{-c(u+v)}(u+v)^{-1}\mathrm{d}u\,\mathrm{d}v = 1/c$: 4.38, 8.76, 10.84 against the computed 4.16, 8.54, 10.62 for $(13, 29)$, $(61, 67)$, $(101, 103)$.

(d) The leading term $\frac{1}{\pi}((\log m)^2 + (\log m')^2)$ of Theorem 1.5(i) is 85–95% of the right-hand side of (i) for $mm' \geq 100$; the difference is the term $c_1 \log Y + c_2$ of Lemma 6.3 at the four cusps, which is still large at $Y \approx 10^2$–$10^3$.

(e) When $mm'$ has several prime factors, the other divisors contribute: for $m = 1$ and $\delta \mid m'$ the cusps with the pair $(m'/\delta, \delta)$ and width $m'/\delta$ see $T(m'/\delta, \delta; n) = r(n/\delta)$ for $n < m'/\delta$, hence contribute about $\frac{1}{\pi\delta}\big(\log\frac{m'}{\delta^2}\big)_+^2$: for $(1, 15015)$ the divisors $\delta = 3, 5, 7, 11, 13$ give 6.72, 2.88, 1.78, 0.82, 0.57 (against 5.84, 2.61, 1.49, 0.67, 0.49 from the formula), which is why this pair has the largest ratio $\|G\|^2/(\log mm')^2$ in Table 1.

In short, $\|G_{m,m'}\|^2$ is, to within a few percent, $\frac{1}{\pi}((\log m)^2 + (\log m')^2) + \frac{5}{\pi^2}\log m\log m'$ plus the contribution of the cusp of denominator 2 and of the other divisors of $mm'$, plus lower-order terms; its ratio to $(\log mm')^2$ tends to $1/\pi$ for $(1, m')$ and to $\frac{1}{2\pi} + \frac{5}{4\pi^2} + \cdots$ for $m \approx m'$ prime, with lower-order terms that are still substantial at $mm' \approx 10^4$. The constant 0.37 of Section 5.3 is the average of these ratios over the sample.

## Data and code

The scripts `mean_square_check.py`, `norm_level20.py`, `certify_modular.py` (arguments `1/2`, `2/3`, `19/20`), `plot_mean_square.py` and `plot_sieve_thresholds.py` (Figures 1 and 2), `local_densities.py` and `cusp_lower_bound.py` (Table 3), the modules of Part I they use (`sup_error_scan.py`, `modular_sieve_scan.py`, `sieve_core.py`) and the outputs `results/mean_square_check_x21.txt`, `results/norm_level20.txt`, `results/certify_modular_output_twothirds.txt`, `results/certify_modular_output_nineteen.txt`, `results/sieve_thresholds.txt`, `results/cusp_lower_bound.txt`, `results/cusp_lower_bound_large.txt` are in the repository https://github.com/Ruqing1963/two-squares-mean-square; the code and data of Part I are at https://github.com/Ruqing1963/two-squares-theta-level. The certification of the sieve parameters uses directed rounding as described in Part I, Appendix A; the computations of Section 5 are in floating point and are evidence, not proof.

## References


[1] A. O. L. Atkin, J. Lehner, *Hecke operators on* $\Gamma_0(m)$, Math. Ann. **185** (1970), 134–160.

[2] C. Bauer, Y. Wang, *On the Goldbach conjecture in arithmetic progressions*, Rocky Mountain J. Math. **36** (2006), 35–66.

[3] R. Chen, *Sums of two squares with few prime factors via theta series of quaternary forms*, preprint (2026), version 3, https://github.com/Ruqing1963/two-squares-theta-level (tag v3.0); version 2 is archived at doi:10.5281/zenodo.23618710. Theorem, lemma and table numbers are the same in versions 2 and 3, except that version 3 adds Theorem 1.2.

[4] H. G. Diamond, H. Halberstam, *A Higher-Dimensional Sieve Method*, with an appendix by W. F. Galway, Cambridge Tracts in Math. **177**, Cambridge Univ. Press, 2008.

[5] H. L. Montgomery, R. C. Vaughan, *The exceptional set in Goldbach's problem*, Acta Arith. **27** (1975), 353–370.

[6] H. Iwaniec, *Topics in Classical Automorphic Forms*, Grad. Stud. Math. **17**, Amer. Math. Soc., 1997.

[7] F. Waibel, *Uniform bounds for norms of theta series and arithmetic applications*, Math. Proc. Cambridge Philos. Soc. **173** (2022), 669–691; arXiv:2012.04311 (equation and lemma numbers refer to the arXiv version).



GUT Geoservice Inc., Montreal, Canada

*Email address*: ruqing@hotmail.com